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system on modules (SoMs). For example, our Xilinx Zynq offering includes six different Zynq devices."

"A development board's native features can be augmented by expansion boards. It should include some kind of expansion, be it FMC, Digilent-compatible peripheral module (Pmod), or both," he adds.

Prasad Reddy, FAE manager, South Asia Pacific, Altera, too feels development boards should have features that accelerate the development time by directly using the board for the final application with a whole slew of application-specific daughter cards. He says, "In addition, manufacturers having a large partner network that also provides application-specific daughter cards is an added advantage."

Resources. The second factor that should be considered is availability of tutorials, reference designs and printed circuit board (PCB) sources. Fletcher informs, "Our development boards come with several tutorials and reference designs based on the latest Xilinx software. PDF copies of all PCB design sources are freely available and the full source may be available through the local Avnet

field application engineers (FAEs)."

After-sales support. The third factor is support, such as online documentation, training and forums. Fletcher says, "These capabilities mean engineers can get the help they need as they are learning new technology and developing their products." Reddy, too, believes, for users adopting new technology, training would be another important factor to consider before buying a board. He says, "Altera offers in-depth training courses that are instructor-lead or online or virtual-classroom based trainings that would offer in-depth insight into the given technology."

Cost. Last, the cost of acquisition also matters. Depending on your budget, you must select the right board that meets your requirements.

Buyers should also consider the quality of documentation and support. Documentation plays a major role in improving user experience, feels Sachin Gupta, product marketing engineer, SoC product line marketing, Microsemi. Hence, he says, "A step towards improvement in hardware kits is to improve documentation quality as well."

There are plenty of offerings out there for FPGA development boards, but surprisingly not all are open and accessible to engineers who want to learn, informs Bowers. He says, "As a student, I was shocked to discover that some major FPGA development board vendors do not include complete schematics of their boards, citing portions of these as proprietary. One vendor did not include any schematics." He adds, "When we built our Mercury FPGA development

The most important factor when looking for a development board is the tools and support offered. If either is lacking, it will make learning how to use the board so much harder

board, we were happy to provide the complete, unabridged schematics, a reference manual and several reference designs to our users. We have also released the entire source code for our USB programmer application so users can modify and use it for their own purposes." He firmly believes that FPGA development boards should be as open and friendly as possible.

Make the right choice

Hardware platforms for FPGAs are primarily influenced by system-level requirements of today's applications' necessities. Vendors are looking to provide low-cost FPGA development boards especially for academic end users. There is continuous effort to bring down the cost of the boards to increase adaptability, informs Gupta. He says, "Ease of use is being ensured by vendors by improving documentation quality and providing an extensive set of demonstration guides and application notes with the kit, which also help in quick learning."

There are many FPGA boards available from individual vendors and third-party companies. Users have a choice of buying a kit that meets their requirements for their proof-of-concept before starting development activity on custom boards. ●

Increasing FPGA densities, board interfaces, supported connectors for add-on modules from third-party vendors and built-in joint test action group (JTAG) interfaces are making boards more powerful and capable